⚠ All documentation must be approved by Wednesday, 25 March 2026

CURRICULUM DESIGN SPECIFICATION · V6

# SASA Applied AI Programme

South African Sugar Association · Mount Edgecombe

6 Sessions | R3,500 / session | R24,150 incl. VAT | 7 hours each | ~11 weeks | In-person · Mount Edgecombe | 23 March 2026

## Programme Overview

**6**
Sessions

**15**
In-person participants

**11**
Online supervisors

**6**
Deliverables

**~2wk**
Cadence between sessions

**R21k**
Excl. VAT

PHASE 1
Sessions 1-3
Data foundations + prompt literacy

PHASE 2
Sessions 4 & build
Technical integration + automation

PHASE 3
Sessions 5-6

Local AI + agents + build day close-out

## Context & Parameters

| | |
|---|---|
| Budget line | Professional Fees (consulting, not training) |
| Location | SASA facility, Mount Edgecombe |
| Session format | 7 hrs: 09:00–12:30 · lunch · 13:30–17:00 |
| Cadence | ~every 2 weeks (~11 weeks total) |
| In-person | 11 managers + 3–5 directors |
| Remote | Up to 11 supervisors |
| AI tool | Claude Pro shared (~R1,700/mo) + Microsoft Copilot |
| Technical scope | Python, Node.js, RAG, large SASA datasets, Ollama, n8n |
| Advanced members | Dave and Sheetal already building — used as benchmarks |
| Paperwork deadline | Wednesday 25 March 2026 |

## Curriculum Philosophy

**Core principle:** Every session produces something usable. The programme is structured so that each session builds on the last — not as abstract theory but as a running consulting engagement with SASA as the client. Participants direct the work; AI assists. The consultant role is to accelerate capability, not to deliver a finished product and leave.

**Prompt engineering thread (S1–3):** Prompt literacy is the foundation the entire programme rests on. Sessions 1, 2, and 3 each contain a dedicated prompt engineering block that progressively builds skill — from first principles (RCTF) through parameterised templates to RAG-aware context management. Every prompt block adds directly to the shared SASA prompt library on SharePoint, building a reusable institutional asset.

**Build day (S6):** The final session is deliberately not a lecture. Teams self-select into build groups based on their automation candidates from Session 4, build something real with senior consultant support, and present it. The programme ends with tools on desks and plans in hand — not a certificate on a wall.

PROMPT TRACK

S1–3

RCTF → reuse → RAG + context

BUILD TRACK

S4–5

Pipeline code → local AI → agents

CLOSE-OUT

S6

Build day → governance → 90-day plans

1

Week 1

Foundations | Prompt Engineering

## Foundations Audit, Strategic Framing & Prompt Engineering

*Mapping SASA's data landscape — building the prompt discipline the whole programme runs on*

The first session maps SASA's current data landscape, standardises tooling, agrees on integration targets, and establishes the prompt foundations every session builds on.

MORNING — AUDIT & SETUP

- Review advanced members' AI work as benchmark
- Map SASA data landscape: R7106, LIMS, Autolab, mill reports
- Configure Claude Pro shared account, Copilot review, setup
- Define programme capstone goals

AFTERNOON — PROMPT ENGINEERING BLOCK

- Role · Context · Task · Format (RCTF) — first template from real SASA report language
- XML tags, 200K context window, system prompts, Projects vs. conversations

- Iteration workflow: draft → test → refine → save
- Common failure modes diagnosed from real examples
- Establish SharePoint prompt library: naming, categories, validation discipline

DELIVERABLE

SASA data & systems map + integration targets + consulting log + seed prompt library (10–15 entries)

2

Week 3

Excel | Reporting | Copilot

## Excel, Reporting & AI-Generated Workflows

*Rebuilding data workflows and eliminating manual report commentary — in one day*

Two sides of the same problem — turning SASA's data into decisions faster — handled in a single session.

### MORNING — EXCEL & DATA WORKFLOWS

- Rebuild one SASA Excel report with Claude co-pilot
- Structured data principles: column naming, type consistency, AI-readable format

EMBEDDED MORNING — PROMPT ENGINEERING BLOCK

- Parameterised templates vs fixed prompts
- Excel prompt library: summarise, flag, extract, compare
- Version discipline — avoiding drift across the team
- Add to shared SASA library

### AFTERNOON — REPORTING WORKFLOWS

- Audit reporting burden — hours/week on interpretive text — ROI conversation
- Narrative workflow: structured data → Claude prompt → draft → sign-off
- Copilot PowerPoint from SASA data summary
- Report commentary template: house style, regulatory language, exception conventions
- One complete AI-assisted report draft reviewed by director in-session

DELIVERABLE

AI-augmented Excel report + exception-flagging template + Excel prompt library + report commentary template + completed AI-assisted report draft

3

Week 5

RAG | Context Management | Large Datasets

## Working with Large Datasets, RAG & Context Management

*Making 10 years of multi-centre SASA data queryable and manageable*

Session 3 addresses SASA's scale challenge — how to query 10 years of operational data without feeding everything to an AI at once.

MORNING — RAG CONCEPTS & DATA WORK

- Introduce RAG — AI-as-librarian pattern, practical analogy, no code required
- Work with SASA multi-year multi-centre data: slice, filter, summarise
- Surface data structural problems that limit AI
- Build simple document index for targeted queries

AFTERNOON — PROMPT ENGINEERING BLOCK

- Context window management: Claude's 200K tokens — paste vs chunk vs RAG
- Chunking strategies: query across chunks without losing coherence
- Summarisation chains: compress 10-year data without losing outliers
- Prompting for synthesis vs extraction
- Multi-document prompting across SASA sources
- Design SASA data-preparation checklist (LIMS, Autolab, mill formats)

DELIVERABLE

Data-preparation checklist + working RAG demonstration + context management reference card

NOTE FOR ONLINE SUPERVISORS

High-value session for online supervisors — RAG concept, context guide, and checklist apply across all roles.

4

Week 7

Python | Node.js | LIMS Pipeline

## Python & Automation Pipelines

*Writing and understanding the code that connects SASA's systems to AI*

Managers direct and review the code — the AI writes it. This session builds the literacy to do that confidently, then puts it to work on a real LIMS data pipeline.

FULL DAY — WHAT WE BUILD TOGETHER

- Consulting frame: you direct and review, AI writes
- Python pipeline: LIMS export → transformation → flagged output → formatted report
- Node.js contrast: Python for data, Node.js for API/SharePoint connectors
- Version control intro for AI-generated code
- Each participant identifies one automation candidate — these form Session 6 build briefs

DELIVERABLE

Working LIMS-to-output pipeline stub + automation candidate list per team area (input to Session 6 build day)

**Pre-session requirement:** Python 3.x, Node.js LTS, and VS Code pre-installed on all attendee machines. IT support required before this session.

5

Week 9

Local AI | Agents | Governance

## Local AI Infrastructure & Agentic Workflows

*The session that changes how SASA thinks about AI — from a tool you ask to a system that acts*

This session deploys SASA's local AI capability — keeping sensitive data on-premise — and introduces agentic automation: AI processes that run without constant human instruction.

MORNING — LOCAL AI DEPLOYMENT

- IT meeting findings mapped against Session 3 data decisions
- Install and configure Ollama + Open WebUI on SASA server with IT present
- Browser access for all team members — no software on individual machines
- Local RAG: index SASA documents, query without internet
- Design data governance framework: cloud vs on-premise — one-page policy for director sign-off

AFTERNOON — AGENTIC WORKFLOWS

- What is an agent? "I ask an AI" vs "the AI does the task and tells me when something needs attention"
- Introduce n8n: open-source, self-hosted, no per-seat licence
- Build SASA agent pipeline: data → exception check → draft alert → SharePoint/file output
- Connect to Session 4 Python output — real data, not a demo
- Design SASA's first agentic use case specification: inputs, triggers, outputs, human review gate

DELIVERABLE

Local AI server operational + working n8n agent pipeline + agentic use case specification + data governance policy

SERVER SPECIFICATION

Minimum: 16 GB RAM · modern CPU · dedicated GPU 4+ GB VRAM · 50 GB SSD · network-accessible
Recommended: 32 GB RAM · RTX 3060 12 GB · 100 GB SSD

6

Week 11

Build Day | Governance | Roadmap

## Build Day, Governance & Programme Roadmap

*Teams build something real — the programme closes with a clear path forward*

The final session is structured in two halves: a morning build day where teams construct a working tool, and an afternoon that turns the programme into a sustained 12-month capability.

MORNING — BUILD DAY

- Teams self-select into 2–3 build groups from Session 4 automation candidate lists

- Build targets: Excel exception report / Python data pipeline / local RAG / n8n agent workflow
- Richard rotates as senior consultant; advanced members lead their groups
- GOAL: every group ships something usable from Monday — not a prototype

AFTERNOON — GOVERNANCE & ROADMAP CLOSE-OUT

- Showcase: each group presents their build (5 minutes each)
- Review all sessions against Session 1 integration targets — what was built, what remains
- Governance framework: acceptable use, data classification, prompt library ownership
- Operating model: who owns what, how new use cases are prioritised
- Skills continuity: who deepens, who needs operational literacy
- 12-month roadmap: 3–5 targets with owners and effort
- Individual 90-day action plans

DELIVERABLE

Team-built working tools (one per group, usable immediately) + data governance policy + 12-month integration roadmap + individual 90-day action plans

## Deliverables Summary

| SESSION | DELIVERABLE |
|---|---|
| S1 | SASA data & systems map + integration targets + consulting log + seed prompt library |
| S2 | AI-augmented Excel report + exception-flagging template + report commentary template + AI-assisted report draft |
| S3 | Data-preparation checklist + RAG demonstration + context management reference card |
| S4 | LIMS-to-output pipeline stub + automation candidate list |
| S5 | Local AI server + n8n agent pipeline + agentic use case spec + data governance policy |
| S6 | Team-built working tools + 12-month roadmap + individual 90-day action plans |

## Extension Programme

### Sessions 7–10 · R3,500 per session

Session 7

**LIMS & Autolab Deep Integration**

AI-assisted exception detection, notification architecture, Autolab data feed

Session 8

**Advanced Agentic Workflows**

Multi-step pipelines, scheduling, error handling, escalation paths

Session 9

**Source System Integration & Production Deployment**

API connectors, live LIMS/SharePoint

Session 10

**Capstone Review, Advanced Governance & Extended Roadmap**

`4 sessions × R3,500 = R14,000 excl. VAT | R16,100 incl. VAT (15%)`

## Prerequisites

BEFORE SESSION 1

- ☐ Claude Pro account set up and shared credentials distributed
- ☐ Admin rights on attendee machines confirmed
- ☐ Projector + video link configured for remote participants
- ☐ SharePoint location created for prompt library
- ☐ Richard reviews Dave & Sheetal's existing AI work before session

BEFORE SESSION 4

- ☐ `Python 3.x` installed on all attendee machines
- ☐ `Node.js LTS` installed on all attendee machines
- ☐ `VS Code` installed — IT support required

BEFORE SESSION 5

- ☐ IT meeting completed — server provisioned and network-accessible
- ☐ AI server meets minimum spec: 16 GB RAM, network-accessible
- ☐ `Ollama` pre-installed and tested on SASA server
- ☐ AI model pre-downloaded (avoid session-day bandwidth issues)
- ☐ `n8n` installed and accessible on server

BEFORE SESSION 6

- ☐ Build briefs confirmed — automation candidates reviewed with Richard
- ☐ Data for build day accessible and prepared in agreed formats

ALL SESSIONS

- ☐ Sample or live SASA data available for the session's focus area

## Investment

| OPTION | SESSIONS | RATE | SUBTOTAL | VAT (15%) | TOTAL |
|---|---|---|---|---|---|
| 6-session programme | 6 × 7 hrs | R3,500 | R21,000 | R3,150 | R24,150 |
| Extension (7–10) | 4 × 7 hrs | R3,500 | R14,000 | R2,100 | R16,100 |
| Full programme (all 10 sessions) | — | — | R35,000 | R5,250 | R40,250 |

- Sessions invoiced individually upon completion
- Payment terms: 14 days from invoice date
- Travel and accommodation invoiced separately where applicable
- Cancellation: 5 business days notice required

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